

Solution Architecture

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Project Name: ShopEZ (E-Commerce Platform)

System Architecture Overview

ShopEZ utilizes a modern, decoupled client-server architecture built entirely on the MERN stack (MongoDB, Express.js, React.js, Node.js), ensuring rapid data flow and modular maintainability.

1. Frontend (Presentation Layer)

- Framework:** Built with React.js 18 and bootstrapped via Vite for incredibly fast build times and Hot Module Replacement (HMR).
- Routing & State:** Utilizes React Router v6 for seamless client-side routing. Global application state (like cart items, user session, and theme preferences) is securely managed via the React Context API.
- Communication:** Interacts with the backend via asynchronous REST API calls using Axios and the native Fetch API.

2. Backend (Application Logic Layer)

- Framework:** Powered by Node.js and Express.js, serving as a highly concurrent RESTful API gateway.
- Security & Middleware:** Implements robust custom middleware to intercept requests, validate JWTs in the Authorization headers, and verify user roles (Customer vs. Admin) before executing controller logic.

3. Database (Data Persistence Layer)

- System:** MongoDB acts as the primary data store, chosen for its flexibility in handling dynamic product attributes and nested order structures.
- ODM:** Mongoose 6.x is employed as the Object Data Modeling library to enforce structured schemas, validate data types, and maintain relationships between Users, Products, Carts, and Orders.

Request Lifecycle Flow

- Client:** The user interacts with the React UI (e.g., clicking "Add to Cart").
- Network:** An HTTP request is dispatched, attaching a secure JWT Bearer token.
- API Gateway:** The Express server receives the request. Middleware validates the JWT.
- Controller:** The specific route controller executes business logic and queries the database via Mongoose.
- Database:** MongoDB processes the query and returns the necessary documents.
- Response:** The server sends a JSON response back to the client.
- UI Update:** React dynamically updates the DOM without reloading the page.